Is K^2 the right GxG kernel under assortative mating?

Derivation and simulation check for `eftedal_reanalysis`

2026-10-14

Question. The models fitted to the Eftedal et al. (2025) correlations write the contribution of pairwise gene–gene interaction (GxG) to the correlation of relatives i, j as $V_{GxG}\, K_{ij}^2$, where K_{ij} is the *assortative-mating (AM) adjusted* additive kernel (e.g. $K = \frac{1+m}{2}$ for siblings). Is this right for additive-by-additive epistasis under primary phenotypic AM at equilibrium?

Answer. It depends on what the interactions are, and for the usual reading of "GxG" it is not.

1. **Random pairwise epistasis** (interaction effects γ_{kl} unrelated to the additive effects β_k): K^2 is *wrong*. AM's inflation of the additive kernel lives entirely in cross-locus genotype correlations aligned with $\beta_k\beta_l$, which GxG does not see. The GxG kernel stays close to the squared pedigree relatedness a_{ij}^2 ($a_{ij} = 2\phi_{ij}$). It gains only a small extra term from sorting on GxG itself, and that term is transmitted with coefficient $\frac{1}{4}$ per meiosis instead of $\frac{1}{2}$. In simulation K^2 overstates the GxG correlation of relatives by 1.7–2.3× for siblings and 2.5–6.4× for second- and third-degree relatives, depending on the strength of AM.

2. **Scale ("aligned") epistasis** (GxG is a nonlinear function of the additive value, e.g. $\gamma_{kl} \propto \beta_k\beta_l$ so $G \propto A^2$): K^2 is *exactly right* when (A_i, A_j) is bivariate normal, i.e. when the curvature is weak. With stronger curvature, AM on the curved scale skews A, and the GxG correlation of relatives then exceeds even K^2.

3. In both cases AM on a phenotype that contains GxG also creates a within-person covariance between the additive and GxG values, and hence cross terms $\mathrm{Cov}[A_i, G_j]$ between relatives. Our model omits them.

4. **Linkage** (Section 6) matters only for interactions between nearby loci, but there it dominates. A tightly linked (cis) pair behaves as one haplotype locus: half of its GxG variance is shared like additive variance (kernel a) and half like dominance. Its kernel is $(a + \delta)/2$ rather than a^2, which is 4× larger for first cousins, already under random mating. AM then scales GxG sharing by a similar factor at every recombination fraction. For cis GxG the true kernel can even exceed K^2 for siblings and more distant relatives. Uniform or main-effect-scaled interactions over all pairs are ≈ 95% unlinked and stay near item 1.

All claims are verified against an allele-level forward simulation (Fig. 1, Table 2). Code is in `theory/gxg_kernel/`: `simulate.py`, `ag_path_model.py` (pathMgr) and `compare.py`.

1 Model

M unlinked biallelic loci; standardized genotypes $x_k = (g_k - 2p_k)/\sqrt{2p_kq_k}$ (base-generation frequencies). For individual i,

$$A_i = \sum_k \beta_k x_{ik}, \qquad G_i = \sum_{k<l} \gamma_{kl}\, x_{ik}x_{il}, \qquad Y_i = A_i + G_i + E_i.$$

Mates are matched on Y with phenotypic correlation ρ_y (primary phenotypic AM), constant across generations, to equilibrium. Base (linkage-equilibrium, random-mating) variances are $V_{A0} = \sum \beta_k^2$ and $V_{G0} = \sum \gamma_{kl}^2$. "Leading order" means as $M \to \infty$ with V_{A0}, V_{G0} fixed (infinitesimal limit). No dominance, no inbreeding, no linkage.

2 Random mating: K^2 with $K = a$

With unlinked loci and no inbreeding, IBD at different loci is independent given the pedigree and $\mathrm{Cov}[x_{ik}, x_{jk}] = a_{ij}$, so

$$\mathrm{Cov}[G_i, G_j] = \sum_{k<l} \gamma_{kl}^2\ \mathrm{E}[x_{ik}x_{jk}]\ \mathrm{E}[x_{il}x_{jl}] = a_{ij}^2\, V_{G0}$$

(Cockerham 1954). Under random mating the GxG kernel is the square of the additive kernel. Under AM the question is *which* quantity gets squared.

3 Assortative mating, random epistasis

3.1 Additive inflation does not reach GxG

The relative-covariance writeup (popstatgenwriteups, `relative_covariance` v1, eq. `same-cross`) shows that at AM equilibrium the same-locus covariance of relatives' genotypes is the random-mating (IBD) value, up to $O(1/M_e)$. Everything AM adds to the additive covariance sits in cross-locus covariances $\mathrm{Cov}[x_{ik}, x_{jl}] \propto \beta_k\beta_l$ $(k \neq l)$. Those are $O(1/M)$ each but $O(M^2)$ in number, which is why the additive kernel inflates from a_{ij} to K_{ij}.

The GxG covariance is $\sum \gamma_{kl}\gamma_{k'l'}\,\mathrm{Cov}[x_{ik}x_{il}, x_{jk'}x_{jl'}]$. When γ is independent of β, a $\beta_k\beta_l$-aligned structure enters only through sums such as $\sum_{kl}\gamma_{kl}^2\beta_k^2 = O(V_AV_G/M)$ or $\sum_l(\sum_k \gamma_{kl}\beta_k)^2 = O(V_AV_G/M)$. These vanish at leading order. So the mechanism that turns a into K for A does nothing for G. If it were the only effect, the GxG kernel under AM would simply be a_{ij}^2.

3.2 What AM does do to GxG: sorting on G itself

Mates are matched on Y, which contains G, so mates' GxG values correlate: $\mu_G \equiv \mathrm{Cov}[G_m, G_f]/V_{GxG}$. Whether that passes to offspring follows from an exact identity. Loci are transmitted independently, and $\mathrm{E}[x_{ok} \mid \text{parents}] = (x_{mk} + x_{fk})/2$, so

$$\mathrm{E}[G_o \mid \text{parents}] = \sum_{k<l} \gamma_{kl}\frac{(x_{mk} + x_{fk})(x_{ml} + x_{fl})}{4} = \tfrac{1}{4}G_m + \tfrac{1}{4}G_f + \tfrac{1}{4}B_{mf}, \tag{1}$$

where $B_{mf} = \sum_{k<l}\gamma_{kl}(x_{mk}x_{fl} + x_{fk}x_{ml})$. B_{mf} is a *cross-parent interaction value*: interactions between one parent's allele at k and the other's at l. To leading order:

- $\mathrm{Var}[B_{mf}] = 2V_{G0}$, and B_{mf} is uncorrelated with every A, G, E in the pedigree. Every covariance it could have with sorted quantities is an odd moment or a β/γ-misaligned sum of order $1/M$. It is shared only by the couple's children.
- The segregation residual $s_G = G_o - \mathrm{E}[G_o \mid \text{parents}]$ has variance $\frac{3}{4}V_{G0}$ and is uncorrelated with s_A. Both depend only on per-locus second moments, which AM leaves at their base values.

So, to leading order, GxG is transmitted *linearly*, like an additive trait but with half the path coefficient and an extra full-sibling-shared term:

$$G_o = \tfrac{1}{4}G_m + \tfrac{1}{4}G_f + \tfrac{1}{4}B_{mf} + s_G, \qquad A_o = \tfrac{1}{2}A_m + \tfrac{1}{2}A_f + s_A.$$

This is a linear path model with one co-path per couple on Y, which is exactly what pathMgr represents. `ag_path_model.py` builds it over a pedigree for each relationship class.

3.3 Equilibrium

With $V_Y = V_A + V_{GxG} + 2c + V_E$ and $c = \mathrm{Cov}[A, G]$ within a person, the co-path gives $\mathrm{Cov}[X_m, Z_f] = \rho_y \,\mathrm{Cov}[X, Y]\,\mathrm{Cov}[Z, Y]/V_Y$, so

$$V_A = \frac{V_A + \rho_y(V_A + c)^2/V_Y}{2} + \frac{V_{A0}}{2}, \qquad V_{GxG} = V_{G0} + \frac{\rho_y(V_{GxG} + c)^2}{7V_Y},$$
$$c = \frac{d}{3}, \qquad d = \mathrm{Cov}[A_m, G_f] = \frac{\rho_y(V_A + c)(V_{GxG} + c)}{V_Y}.$$

Equivalently $V_{GxG} = V_{G0}/(1 - \mu_G/7)$, against $V_A = V_{A0}/(1 - m)$ for the additive part: AM inflates GxG variance about seven times more weakly. In the stationary pathMgr model the descendants' variances reproduce the founders' exactly, which checks this fixed point.

3.4 Resulting kernels

Two classes have closed forms in GxG units:

$$\text{parent–offspring: } K_G = \frac{1 + \mu_G}{4}, \qquad \text{full siblings: } K_G = \frac{1 + \mu_G}{8} + \frac{V_{G0}}{8V_{GxG}} = \frac{1}{4} + \frac{3\mu_G}{28}.$$

Compare $K^2 = \left(\frac{1+m}{2}\right)^2 = \frac{1}{4} + \frac{m}{2} + \frac{m^2}{4}$. At random mating ($\mu_G = 0$) every class reduces to a^2, which pathMgr reproduces exactly. Table 1 gives all classes at illustrative, test-score-like parameters. The AM increment over a^2 is small for close relatives and negligible for in-laws, whereas K^2 inflates steadily with distance.

3.5 The A–GxG cross term

Because mates are matched on $A + G$, $\mathrm{Cov}[A_m, G_f] = d \neq 0$, and offspring inherit a within-person covariance $c = d/3$. Relatives then also covary through $\mathrm{Cov}[A_i, G_j]$, contributing $\mathrm{Cov}[A_i, G_j] + \mathrm{Cov}[G_i, A_j]$ to the phenotypic covariance. At $\rho_y = 0.45$ this cross term is 0.05–0.07 in correlation units for first-degree relatives, as large as or larger than the AM increment in K_G itself (Table 1). Neither our model nor Eftedal's has it.

class	a^2	$\rho_y = 0.30$			$\rho_y = 0.45$		
		K_G	K_A^2	cross	K_G	K_A^2	cross
parent-offspring	0.2500	0.2653	0.3210	0.045	0.2731	0.3675	0.070
full sibs	0.2500	0.2566	0.3210	0.030	0.2599	0.3675	0.047
half sibs	0.0625	0.0713	0.1066	0.030	0.0766	0.1445	0.049
grandparent	0.0625	0.0715	0.1034	0.033	0.0773	0.1359	0.054
avuncular	0.0625	0.0686	0.1034	0.024	0.0721	0.1359	0.039
first cousins	0.0156	0.0186	0.0335	0.015	0.0207	0.0508	0.025
half cousins	0.0039	0.0055	0.0112	0.009	0.0070	0.0203	0.017
second cousins	0.0010	0.0016	0.0035	0.005	0.0022	0.0071	0.009
step sibs	0.0000	0.0065	0.0032	0.019	0.0121	0.0125	0.037
co-cousins	0.0000	0.0006	0.0002	0.003	0.0012	0.0010	0.006

Table 1: GxG kernels at base $V_{A0} = 0.40$, $V_{G0} = 0.20$, $V_E = 0.40$ (random epistasis; pathMgr path model). K_G: correct kernel; K_A^2: the current model's kernel built from the AM-adjusted additive kernel K_A; a^2: squared pedigree relatedness. "cross" is $(\mathrm{Cov}[A_i, G_j] + \mathrm{Cov}[G_i, A_j])/(2\sqrt{V_A V_{GxG}})$, the A–GxG cross term our model omits.

4 When K^2 is right: scale epistasis

If interaction effects follow the additive ones, $\gamma_{kl} = \lambda\beta_k\beta_l$, then $G = \frac{\lambda}{2}(A^2 - \sum_k \beta_k^2 x_k^2)$. Up to an $O(1/M)$ term this is a quadratic function of the additive value, i.e. a nonlinear scale on A. For mean-zero bivariate-normal (A_i, A_j), $\mathrm{Cov}[A_i^2, A_j^2] = 2\,\mathrm{Cov}[A_i, A_j]^2$ (Isserlis) and $\mathrm{Var}[A^2] = 2V_A^2$, so

$$\frac{\mathrm{Cov}[G_i, G_j]}{V_{GxG}} = K_{ij}^2 \quad \text{exactly, AM included.}$$

More generally, for any smooth scale $f(A)$, $\mathrm{Cov}[f(A_i), f(A_j)] = \sum_n c_n^2\, n!\, (K_{ij}V_A)^n$ (Hermite expansion), and K^2 is its second-order term. So the current kernel is the correct kernel for *non-additivity of scale*: a phenotype that is a curved function of a single additive liability. It is not the kernel for interactions between independently acting loci.

Caveat: AM on a curved scale distorts A. The identity needs (A_i, A_j) bivariate normal. AM on $Y = A + \lambda A^2/2 + E$ breaks this when the curvature matters. Matching on Y assorts more precisely where Y is steeper in A (the upper tail for $\lambda > 0$), so over generations the upper tail spreads more than the lower. A becomes skewed, and it correlates with its own square, and hence with G. In simulation (`results/*_curved`) the skewness of A reaches 0.8 after 12 generations of strong AM, and the within-person correlation of A and G about 0.4. The GxG kernel then exceeds K_A^2: full siblings 0.72 vs 0.49, 0.85 vs 0.64; first cousins 0.41 vs 0.13, 0.70 vs 0.31 (moderate, strong AM). It also carries a large A–GxG cross term. For a strongly curved scale, K^2 is therefore a lower bound, and the cross term cannot be ignored.

5 Simulation check

`simulate.py`: N = 20,000 per generation, M = 1,000 unlinked loci, 20 generations of rank-matching AM on Y, two children per couple, and a second assortative union for 30% of the last parental generation (half and step siblings). Three replicates per setting.

- Random epistasis: 4,000 random locus pairs, base $V_{A0} = 0.45$, $V_{G0} = 0.30$, $V_E = 0.25$.
- Scale epistasis: $G \propto A^2$. The identity test uses negligible curvature ($V_{G0} = 10^{-4}$; the kernel ratio does not depend on the size of λ); the curvature caveat uses $V_{G0} = 0.03$ (`*_curved`).
- AM: strong ($\rho_y \approx 0.86$) or moderate ($\rho_y \approx 0.64$).

Predictions are evaluated at each replicate's measured base variances and ρ_y. The equilibrium V_{GxG} and c match the fixed point (e.g. strong: $c = 0.069$–0.070 simulated vs 0.067–0.074 predicted).

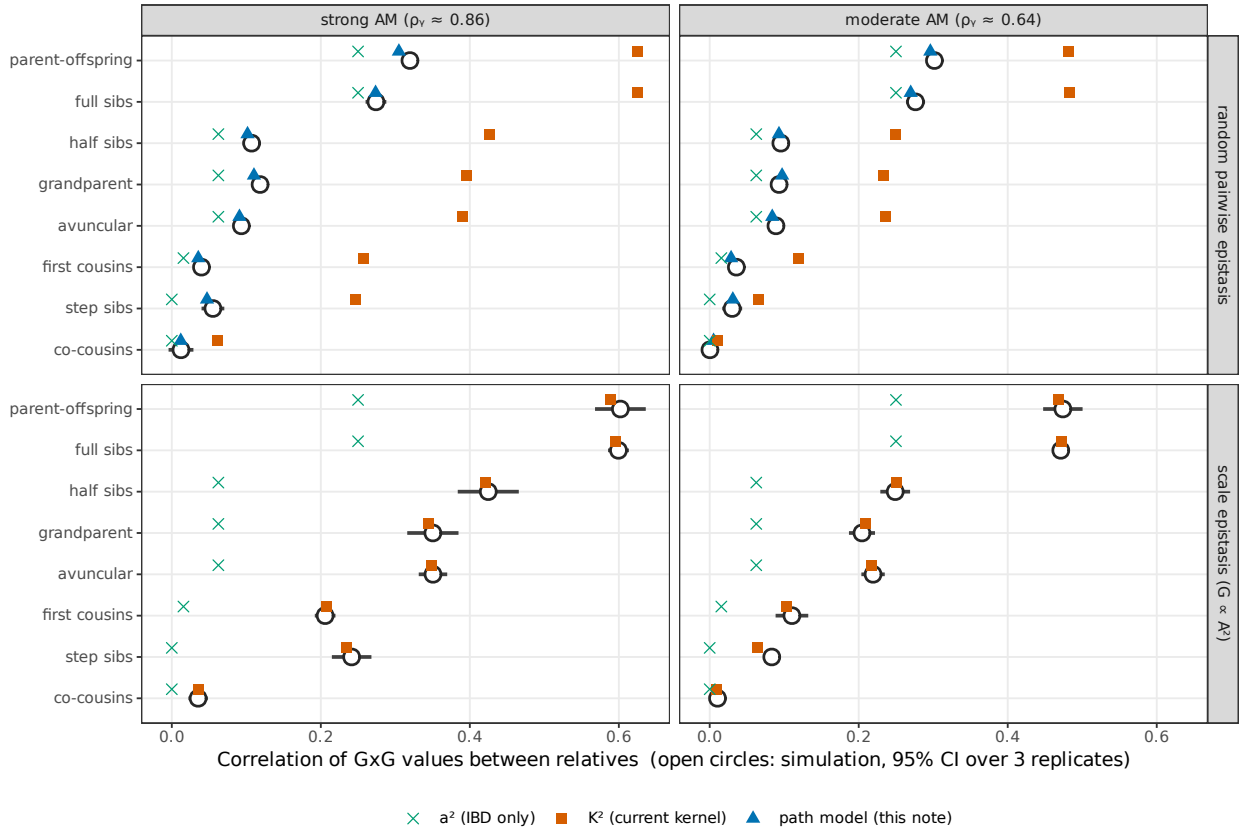


Figure 1: Correlation of GxG values between relatives: simulation (open circles, 95% CI over replicates) against the path model of this note, the current kernel K^2, and a^2.

Result. **Random epistasis:** the path model matches every class in both AM regimes to within 0.01, with one exception. Mother–offspring under strong AM is 0.320 simulated vs 0.305 predicted, and the gap does not shrink with $M = 3{,}000$ loci. It comes from an asymmetry of the simulated mating scheme, which adds matching noise to the female's score only. In a diagnostic run $\mathrm{Cov}[B_{mf}, G_m]/V_{GxG} = +0.024$ and $\mathrm{Cov}[B_{mf}, G_f]/V_{GxG} = -0.019$: zero in sum, as the theory requires, but not separately. Classes that depend on both parents symmetrically (e.g. full siblings, 0.274 vs 0.273) are unaffected. The measured $\mathrm{Var}[B_{mf}] = 2.01\, V_{G0}$ matches the derivation. K^2 is too large everywhere. Under moderate / strong AM it overstates siblings 1.7× / 2.3×, avuncular 2.7× / 4.2× and first cousins 3.3× / 6.4×, and step siblings 2.2× / 4.5×. **Scale epistasis:** with

class	random epistasis sim (SE)	path model	K_A^2	a^2	scale epistasis sim (SE)	K_A^2
parent-offspring	0.320 (0.005)	0.305	0.626	0.250	0.602 (0.017)	0.589
full sibs	0.274 (0.007)	0.273	0.625	0.250	0.600 (0.007)	0.595
half sibs	0.107 (0.006)	0.102	0.427	0.062	0.425 (0.021)	0.421
grandparent	0.118 (0.005)	0.110	0.395	0.062	0.350 (0.018)	0.344
avuncular	0.093 (0.002)	0.091	0.390	0.062	0.351 (0.010)	0.349
first cousins	0.040 (0.003)	0.036	0.257	0.016	0.206 (0.007)	0.207
step sibs	0.055 (0.008)	0.047	0.247	0.000	0.241 (0.014)	0.235
co-cousins	0.012 (0.009)	0.012	0.062	0.000	0.035 (0.007)	0.036

Table 2: Simulated GxG kernel vs predictions (strong AM; moderate AM in `results/comparison.csv`). Random-epistasis rows compare with the path model; scale-epistasis rows with K_A^2 computed from the simulated additive kernel.

negligible curvature the simulated GxG kernel matches K_A^2, built from the simulated additive kernel. Under strong AM the ratio is 0.99–1.03 in every class with $K_A^2 > 0.02$. Under moderate AM every class agrees within 0.01, except step siblings (0.083 vs 0.064, about 5 replicate SEs from 3 replicates).

6 Linked interacting loci

Everything above assumes unlinked loci. Here the interacting loci sit on a genetic map, and the pair (k, l) has recombination fraction $c_{kl} \in [0, \frac{1}{2}]$. Code: `digenic.py` (exact random-mating coefficients by gene dropping), `ag_linked_model.py` (pathMgr model), `simulate_linked.py` (haplotype-level simulation) and `compare_linked.py`.

6.1 Cis and trans parts of a pair

Write the centred allele values on an individual's two haplotypes as h_{1k}, h_{2k}, so $x_k = h_{1k} + h_{2k}$ and

$$x_k x_l = \underbrace{h_{1k}h_{1l} + h_{2k}h_{2l}}_{\text{cis}} + \underbrace{h_{1k}h_{2l} + h_{2k}h_{1l}}_{\text{trans}}.$$

At linkage equilibrium the two parts are uncorrelated and each carries half of the pair's variance. A meiosis passes on one of the parent's cis products with probability $1 - c$ and one of its trans products with probability c. A product across the two parental gametes becomes a trans product in the child. For $c = \frac{1}{2}$ this is the unlinked transmission of Section 3. For $c = 0$ the pair is a single locus whose alleles are haplotypes: the cis part is that locus's *additive* value and the trans part its *dominance* deviation.

6.2 Random mating

The pair contributes $\gamma_{kl}^2\, \psi_{ij}(c_{kl})$ to $\mathrm{Cov}[G_i, G_j]$, where $\psi_{ij}(c) = \mathrm{E}[a_{ij}(k)\, a_{ij}(l)]$ is the expected product of the realized relatedness at the two loci, the digenic identity coefficient (Cockerham 1956; Weir & Cockerham 1977). With $s = (1-c)^2 + c^2$, the probability that two meioses of one

parent are concordant at the two loci,

$$\psi_{\mathrm{PO}} = \tfrac{1}{4}, \quad \psi_{\mathrm{FS}} = \tfrac{s}{4} + \tfrac{1}{8}, \quad \psi_{\mathrm{HS}} = \tfrac{s}{8}, \quad \psi_{\text{lineal},\, n \text{ meioses}} = \tfrac{1}{4}\left(\tfrac{1-c}{2}\right)^{n-1}, \qquad \psi(0) = \tfrac{a+\delta}{2}, \quad \psi(\tfrac{1}{2}) = a^2,$$

with δ the probability that both alleles are IBD ($\frac{1}{4}$ for full siblings, 0 for the other classes here). Table 3 gives gene-dropping values. Parent–offspring is unaffected, because a parent shares exactly one allele with a child at every locus. For every other class linkage increases GxG sharing, and the increase grows with the distance of the relationship: complete linkage multiplies first-cousin sharing by 4 and second-cousin sharing by 16. Fully linked GxG is half additive-like (kernel a) and half dominance-like (kernel δ). A variance-components model will therefore absorb it into V_A and the sibling excess, not into an a^2-shaped term.

Table 3: Random-mating GxG kernel $\psi(c)$ for a pair of loci at recombination fraction c (gene dropping, 4×10^5 draws; agrees with the closed forms above to ± 0.0005).

class	a	$c = 0$	0.01	0.1	0.5 ($= a^2$)	$\psi(0)/a^2$
parent-offspring	0.5000	0.2500	0.2500	0.2500	0.2500	1
full sibs	0.4998	0.3748	0.3697	0.3294	0.2500	2
half sibs	0.2500	0.1250	0.1225	0.1023	0.0625	2
grandparent	0.2505	0.1253	0.1240	0.1127	0.0627	2
avuncular	0.2497	0.1248	0.1217	0.0984	0.0624	2
first cousins	0.1248	0.0624	0.0600	0.0418	0.0157	4
half cousins	0.0625	0.0312	0.0300	0.0207	0.0040	8
second cousins	0.0313	0.0157	0.0147	0.0085	0.0010	16
step sibs	0.0000	0.0000	0.0000	0.0000	0.0000	–
co-cousins	0.0000	0.0000	0.0000	0.0000	0.0000	–

6.3 Assortative mating

Per recombination bin b (fraction c_b, base variance V_{b0}), the transmission is linear to leading order, as in Section 3:

$$G^{\mathrm{cis}}_{b,o} = \tfrac{1-c_b}{2}\left(G^{\mathrm{cis}}_{b,m} + G^{\mathrm{cis}}_{b,f}\right) + \tfrac{c_b}{2}\left(G^{\mathrm{trans}}_{b,m} + G^{\mathrm{trans}}_{b,f}\right) + s_{cb}, \qquad G^{\mathrm{trans}}_{b,o} = \tfrac{1}{4}B_{b,mf} + s_{tb},$$

with $\mathrm{Var}[B_b] = 2V_{b0}$, $\mathrm{Var}[s_{cb}] = \frac{V_{b0}}{2} - \frac{V_{b0}}{4}\left((1-c_b)^2 + c_b^2\right)$ and $\mathrm{Var}[s_{tb}] = \frac{3}{8}V_{b0}$. Mates are matched on $Y = A + \sum_b (G^{\mathrm{cis}}_b + G^{\mathrm{trans}}_b) + E$. `ag_linked_model.py` solves the equilibrium of this system and builds the stationary pathMgr model for each class. At $\rho_y = 0$ it reproduces the gene-dropping ψ exactly, and at $c = \frac{1}{2}$ it reproduces the unlinked model of Section 3 to machine precision. Table 4 uses the illustrative parameters of Table 1, with all interacting pairs at one c.

1. **Linkage and AM roughly multiply.** At $\rho_y = 0.45$, AM raises GxG sharing by a similar factor at every c (first cousins $\times 1.29$ at $c = 0$, $\times 1.33$ at $c = \frac{1}{2}$). Linkage is the larger effect for anyone beyond first-degree relatives.

2. **The cis part does inherit sorting, as the additive value does, but only its own share.** At $c \approx 0$ half of GxG is transmitted with coefficient $\frac{1}{2}$, like A, so mate correlation on it accumulates across generations. That mate correlation comes from GxG's own contribution to Y (order $\rho_y V_{GxG}/V_Y$), however, not from the β-aligned cross-locus structure that inflates K_A. The increment is therefore modest.

3. K^2 **is no longer an upper bound.** Tightly linked GxG grows like $a/2$ rather than a^2. At $\rho_y = 0.45$ and $c = 0$ its kernel exceeds K_A^2 for full siblings (0.396 vs 0.368), half siblings (0.150 vs 0.145), grandparents (0.156 vs 0.136) and first cousins (0.081 vs 0.051). Only parent–offspring stays below K_A^2; even in-laws exceed it, because their K_A^2 is tiny. This mirrors Section 3, where unlinked GxG is far *below* K^2 for every class.

4. V_{GxG} barely changes with c (0.203–0.207), and the A–GxG cross term grows mildly with linkage.

Table 4: GxG kernel with all interacting pairs at recombination fraction c; base $V_{A0} = 0.40$, $V_{G0} = 0.20$, $V_E = 0.40$ (linked path model). RM: random mating.

	$c = 0$		$c = 0.1$		$c = \frac{1}{2}$	
class	RM	ρ_y=0.45	RM	ρ_y=0.45	RM	ρ_y=0.45
parent-offspring	0.2500	0.2877	0.2500	0.2822	0.2500	0.2731
full sibs	0.3750	0.3960	0.3300	0.3468	0.2500	0.2599
half sibs	0.1250	0.1498	0.1025	0.1233	0.0625	0.0766
grandparent	0.1250	0.1561	0.1125	0.1381	0.0625	0.0773
avuncular	0.1250	0.1513	0.0985	0.1186	0.0625	0.0721
first cousins	0.0625	0.0809	0.0418	0.0549	0.0156	0.0207
half cousins	0.0312	0.0452	0.0208	0.0307	0.0039	0.0070
second cousins	0.0156	0.0250	0.0085	0.0142	0.0010	0.0022
step sibs	0.0000	0.0177	0.0000	0.0157	0.0000	0.0121
co-cousins	0.0000	0.0054	0.0000	0.0035	0.0000	0.0012

6.4 Effect distributions

GxG on a real genome is a mixture over c: genome-wide pairs are mostly unlinked (two loci fall on the same chromosome with probability $\sum_i L_i^2/(\sum_i L_i)^2 \approx 5\%$ for human autosomes, and are tightly linked far more rarely), whereas cis interactions within a gene or regulatory domain have $c \approx 0$. Which distribution holds is a question about the biology of γ_{kl}:

(1) **Uniform**, γ_{kl} i.i.d. over all pairs. GxG variance is $\approx 95\%$ unlinked, so the kernel is a^2 up to a few percent.

(2) **Scaled by main effects**, $\gamma_{kl} = \beta_k\beta_l\delta_{kl}$ with δ_{kl} i.i.d. The *mean* of δ matters. With $\mathrm{E}[\delta] = 0$ the signs are unrelated to the additive effects and the result is the same as (1); the weighting only redistributes variance among pairs. With $\mathrm{E}[\delta] = \mu \neq 0$, G contains $\frac{\mu}{2}\big(A^2 - \sum_k \beta_k^2 x_k^2\big)$: genome-wide *scale* epistasis (Section 4), for which the AM-adjusted K^2 is right.

(3) **Scaled by main effects and proximity.** A weight $(1-c_{kl})$ does not localize the interactions. Because $c \leq \frac{1}{2}$, it lies in $[\frac{1}{2}, 1]$, so unlinked pairs keep a quarter of the variance weight of the tightest pairs and vastly outnumber them. In the simulation it moves the unlinked share of GxG variance only from 95% to 88%. A proximity weight must vanish for unlinked pairs: $1 - 2c_{kl} = e^{-2d_{kl}}$ (Haldane, d in Morgans), or more generally $e^{-d_{kl}/\lambda}$ on the same chromosome with a biological length scale λ. We use $\lambda = 0.1\,\mathrm{cM}$ (within a gene, $\sim$100 kb) and $\lambda = 5\,\mathrm{cM}$ ($\sim$5 Mb, across neighbouring genes).

(3d) **Directional and local**, $\gamma_{kl} = \beta_k\beta_l e^{-d_{kl}/\lambda}$ (no random sign): scale epistasis *within* genes.

Simulation. 20 chromosomes of 1 M, each with 25 genes of 4 loci spread over 0.1 cM ($M = 2000$); Haldane map; $N = 20{,}000$; $V_{A0} = 0.45$, $V_{G0} = 0.30$, $V_E = 0.25$; 20 generations of random mating or strong AM ($\rho_y \approx 0.86$); three replicates; relative classes and measures as in Section 5. Predictions use each architecture's realized distribution of GxG variance over c (10 bins). Within-chromosome AM disequilibrium builds up on a time scale of $1/c$ generations, so for tight pairs the simulation is still approaching equilibrium at generation 20 ($V_A = 1.16$ against an equilibrium value of 1.32 for architecture 3c). We therefore compare with the path model iterated for the same number of generations from the linkage-equilibrium base. That model gives $V_A = 1.16$, $V_{GxG} = 0.38$, against the simulation's 1.16, 0.38. The equilibrium predictions are 0.02–0.03 higher for the tight architectures. Figure 2. Across the five undirected architectures, both regimes and all eight classes, the linkage path model is within 0.016 of the simulated GxG correlation (median absolute error 0.0037; 6 of 80 cells beyond 3 replicate standard errors, which are themselves estimated from three replicates). K^2 overstates the sibling, half-sibling and first-cousin GxG correlation under strong AM by 1.3–6.9×. For the directional within-gene architecture (3d) under strong AM the path model, which assumes interaction effects unrelated in sign to the additive effects, gives full sibs 0.508 vs. simulated 0.520, half sibs 0.306 vs. simulated 0.309, first cousins 0.228 vs. simulated 0.233.

Why local directional epistasis behaves like random-sign epistasis. Architectures 3c and 3d give the same kernels within replicate noise, and both match the path model, which uses only the variances. AM's β-aligned cross-locus covariances, $\mathrm{Cov}[x_k, x_m] \propto \beta_k \beta_m$, enter $\mathrm{Cov}[G_i, G_j]$ through products over *pairs of pairs*, $\sum \gamma_{kl}\gamma_{mn}\, C_{km} C_{ln}$. When directional interactions cover all $O(M^2)$ pairs (genome-wide scale epistasis) these terms add up to an $O(1)$ effect, and they are what make K^2 exact in Section 4. When the directional pairs are local, only $O(M)$ of them exist, and the sum is $O(1/M)$. Directionality matters for the kernel only when the nonlinearity acts on a genome-wide aggregate.

7 Implications for the reanalysis

1. **What the fitted "GxG" term currently means.** With kernel K^2 it is a *scale / nonlinearity* component (a curved function of the additive value), not interaction among independent loci. That is a legitimate and arguably plausible model for test scores (floor and ceiling effects), but it should be labelled as such.

2. **For classical random GxG, replace K^2 by K_G.** A simple and adequate approximation is a_{ij}^2, which ignores sorting on GxG: at $\rho_y \approx 0.45$ it errs by ≤ 0.023 for first-degree relatives and by ≤ 0.015 for every other class in Table 1. With a^2, GxG contributes 1 to MZ twins, $\frac{1}{4}$ to DZ twins, full siblings and parent–offspring, $\frac{1}{16}$ to second-degree relatives (half siblings, avuncular, grandparent), $\frac{1}{64}$ to first cousins, and essentially nothing to more distant relatives or in-laws. It becomes almost a "close-relative excess" term, identified mainly by the MZ/DZ contrast. Expect it to compete much less with the family-environment term and with V_A.

3. **UZ-linked classes.** The MZ/DZ mixture should be applied to the squared kernel $(p\, a^2_{\mathrm{MZ}} + (1-p)\, a^2_{\mathrm{DZ}})$, as the code already does for K^2.

4. **The A–GxG cross term** is at least as large as the AM correction to K_G. Ignoring both (the a^2 kernel) is consistent; including only one of them is not.

5. Suggested follow-up: refit the core models with GxG kernel a^2 ("random GxG") next to K^2 ("scale"), and compare by block cross-validation.

6. **Cis GxG is not separately identifiable from these data.** Its kernel $(a+\delta)/2$ is (exactly at random mating, approximately under AM) a linear combination of the additive kernel and the sibling (dominance-like) excess, so in the fitted models it loads on V_A and on whatever term carries the sibling excess. The "GxG" term with kernel a^2 (or K^2) speaks only to interactions between unlinked loci, which are the bulk of genome-wide pairs.

Caveats. Sections 2–6 assume unlinked loci (Section 6 relaxes this; there the AM equilibrium for very tight pairs is approached on a time scale of $1/c$ generations, and the simulation runs 20); the infinitesimal limit; interaction effects independent of additive effects (case 1) or exactly aligned with them (case 2), with real architectures plausibly a mixture, whose kernel is the corresponding mixture; primary phenotypic AM at equilibrium; no dominance.



Figure 2: Simulated GxG correlation between relatives (open circles, 95% CI over three replicates) with linked interacting loci, by effect architecture (rows) and mating regime (columns), against the linked path model, the current kernel K_A^2, and a^2.